



Equation List for Exam

The listed equations serve the purpose of preparation for the calculation part of the aircraft design exam and should be understood and learned by heart. Exam candidates must also be able to derive further relations from these equations.

Chapter 2: Aerodynamics

2.1	Wing geometrical definitions	$AR =$ $\lambda =$ $CMAC =$
2.1	Lift, drag and moment	$L =$ $D =$ $M =$
2.1	Lifting line theory, Polhamus method, Slender body theory	Lifting line theory: $C_{L\alpha} =$ Polhamus method: $C_{L\alpha} =$ with β (subsonic) Slender body theory: $C_{L\alpha} =$



2.2	Components of drag coefficient	$C_{D0} =$
2.2	Equivalent aircraft & component parasite drag area	
2.2	Reynolds number	$Re = \underline{\hspace{2cm}}$
2.2	Component drag coefficient and parasite drag area	$C_{D0,j} =$ $f_j =$
2.2	Lift-induced drag coefficient	$C_{Di} =$ With $k = \underline{\hspace{2cm}}$ Or $k = \underline{\hspace{2cm}}$
2.6	Stability measure	
2.6	Aerodynamic center of wing or horizontal tail	$x_{AC,W} = \underline{\hspace{4cm}}$ Or $x_{AC,HT} = \underline{\hspace{4cm}}$



2.6	Neutral point calculation	$x_{AC} =$ $C_{L\alpha, W} x_{AC} =$
2.6	Trim around lateral axis	$C_{M(x=0)} =$

Chapter 3: Structural design

3.1	Weight breakdown of take-off and landing weight	
3.3	Growth factor for payload mass (example)	
3.4	Vertical load factor	$n_z = \underline{\hspace{2cm}}$
3.4	Load factors during maneuvers	$n_z = \underline{\hspace{2cm}}$ $n_z = \underline{\hspace{2cm}}$



3.4	Conversion from TAS to EAS	$V_{EAS} V_{TAS} =$
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Chapter 4: Propulsion design

4.2	Engine thrust	
4.3	Thrust and power specific fuel consumption	$TSFC =$ $PSFC =$

Chapter 5: Point performance

5.2	Specific excess power	$SEP = \underline{\hspace{2cm}}$
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Chapter 6: Mission performance

6.3	Weight reduction rate	$dw =$
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6.3	Flight condition I: duration and range	$\Delta t = \text{—————}$ $\Delta s = \text{—————}$ Turbojet/turbofan: $SFC_x = TSFC_x; n_V \approx 0$ Turboprop: $SFC_x = PSFC_x; n_V \approx -1$
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Chapter 7: Design chart

7.1	Definition of thrust-to-weight ratio	$\frac{T}{w} \text{ —————} = [-]$
7.1	Definition of wing loading	

General

	Lift coefficient, lift curve slope and angle of attack	$C_L =$
	Mach number	$M = \text{—}$